

PERSONAL LEARNING JOURNEY

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Assigned role: Modelling, Evaluation and Interpretation

My contribution

My job was to train the models, compare them fairly, choose the final one and explain what it learned. I compared six models: Logistic Regression, Naive Bayes, k-Nearest Neighbours, a Decision Tree, a Random Forest and Gradient Boosting. I also used a simple baseline that always predicts the most common result. Later, I also tested more advanced models, including XGBoost, a neural network and support vector machines. I prepared the comparison tables, the confusion matrix and the charts that show which features matter most.

What I learned

I learned that accuracy can be misleading. One model was right 54.0% of the time, which sounds good, but it almost never predicted a draw. About one match in four is a draw, so that model was not useful. We therefore used Macro-F1, a score that gives home wins, draws and away wins equal importance. I also checked log loss, which shows whether the model's probabilities can be trusted. Finally, I learned to test models in time order, just like predicting real future matches. We trained on seasons up to 2022, checked on 2023 and tested on 2024 onwards.

Challenge

Draws were the biggest problem: without any adjustment, four of the six models almost never predicted a draw. The scores did not always agree either. Naive Bayes scored well on Macro-F1 (0.473), but its probabilities were poor (log loss 2.35, more than twice that of the other models). Another challenge was self-discipline. The test seasons must be used only once, at the very end. Looking at them while choosing a model would make our result look better than it really is.

Solution

I told the models to pay more attention to draws, using a setting called class weighting. This raised the draw score (Draw F1) from almost zero to between 0.27 and 0.34. The Random Forest had the best Macro-F1 (0.492) and also trustworthy probabilities, so we chose it as the final model. On the test seasons, used only once, it scored 0.473 Macro-F1 and was right 48.9% of the time, against 0.202 and 43.4% for the simple baseline. Before testing the extra models, I set a rule: replace the Random Forest only if a new model is better by more than 0.01. None of them was.

Reflection

The best models of very different types (bagging, boosting and kernel methods) all reached about the same score, around 0.49 on the validation season. This tells me the limit is the information available before kickoff, not the choice of algorithm. Football is hard to predict: even the team with the stronger starting eleven loses about 3 in 10 matches that are not draws. I learned that an honest, carefully tested result is worth more than a high score that comes from peeking at the test data.

Future improvement

I would check whether the model's probabilities are well calibrated, which we did not test. I would also try a smarter tuning method (Bayesian optimisation) for larger models, and compare our predictions with bookmakers' odds. Most of all, I would look for features that help to predict draws, which are still the hardest result (test Draw F1 0.326).